



**College of Architecture and Planning  
Department of Building Engineering**

## **Receiver Analyzing Report**

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# 1. Introduction: Acoustic Evaluation of the Hospital Ward

The acoustic environment of a healthcare facility is a critical factor in both patient recovery and staff performance. High noise levels and excessive reverberation can lead to increased patient stress and a higher risk of medical errors due to the "acoustic masking" of verbal instructions.

## 1.1 Project Scope

This report evaluates the acoustic characteristics of a specific hospital ward with a total volume of **2173.00 m<sup>3</sup>**. The analysis is based on simulated sound decay across three specific receiver points to determine the space's clinical suitability.

## 1.2 Objectives

Based on the provided simulation data, the objectives of this report are as follows:

- **Quantify Reverberation Time  $T_{30}$ :** To measure the duration of sound decay across the frequency spectrum at three strategic locations.
- **Assess Spectral Balance:** To identify specific frequency imbalances, particularly the "boomy" low-frequency energy observed.
- **Evaluate Communication Clarity:** To determine if the reverberation at **1000 Hz** (averaging between 0.84s and 1.18s) allows for sufficient speech intelligibility.
- **Verify Compliance:** To evaluate the integrated decay curves against the **NC 30** noise criteria requirements for healthcare environments.

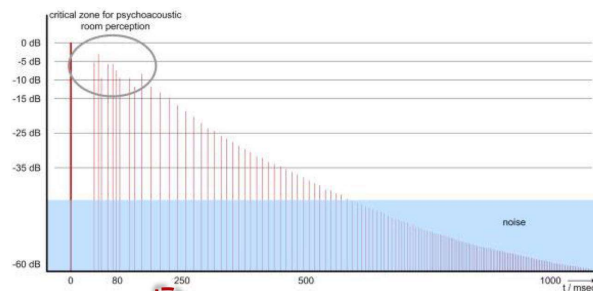
## 2. Reverberation Time: Criteria and Definitions

Reverberation time is the primary metric used to quantify the "liveliness" or "echoic" nature of a space<sup>8</sup>. It is formally defined as the time required for the sound pressure level to decay by 60 dB after a steady-state sound source has been abruptly terminated<sup>7</sup>.

## 2.1 Key Performance Metrics

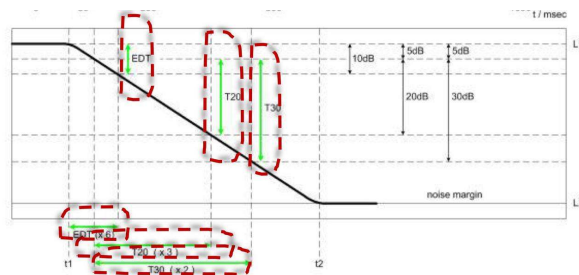
In professional acoustic analysis, Reverberation time is often evaluated through specific decay intervals to ensure measurement stability:

- **$T_{30}$ :** This is the time calculated by measuring the decay from -5 dB to -35 dB and extrapolating the slope to the full 60 dB range<sup>6</sup>, shown in **Figure 1**. It is widely regarded as the most reliable indicator of a room's objective reverberation<sup>5</sup>.
- **Early Decay Time (EDT):** Calculated from the first 10 dB of decay, this metric relates more closely to the subjective human perception of "reverberance" than the standard  $T_{30}$ <sup>1</sup>. Shown in **Figure 1**.



**Figure 1.** illustrates the **impulse response** of a room.

- Decay Curve:** A graphical representation of energy versus time shown in **Figure 2**. A smooth, linear curve indicates a diffuse sound field, while fluctuations may indicate distinct echoes or non-uniform absorption <sup>3</sup>.

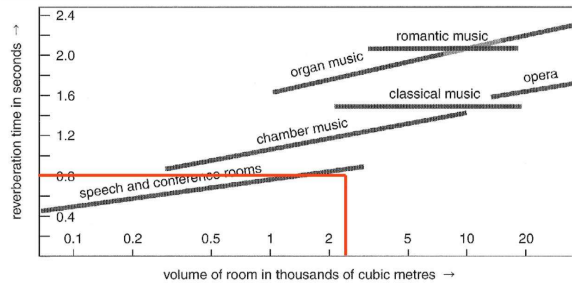


**Figure 2.** sound decay curve.

## 2.2 Acoustic Requirements for Healthcare:

Acoustic comfort in hospitals is critical for patient recovery and staff performance. Standards such as the Health Technical Memorandum and BB93 provide the following guidelines:

- **Target Reverberation time Values:** For multi-bed wards, the recommended reverberation time is typically between 0.8 and 1.0 seconds at mid-frequencies <sup>4</sup>. Times exceeding this range can lead to "acoustic fatigue" and decreased speech intelligibility



**Figure 3.** relationship between room volume and the **optimum reverberation time** for different acoustic functions.

- **Low-Frequency Management:** High reverberation at low frequencies (below 250 Hz) creates a "boomy" environment that can mask the high-frequency transients of medical alarms, posing a safety risk <sup>1</sup>.
- **Speech Intelligibility:** Excessive reverberation degrades the Speech Transmission Index (STI), making it difficult for medical staff to communicate critical instructions, especially in high-stress environments <sup>2</sup>.

### 3. Methodology:

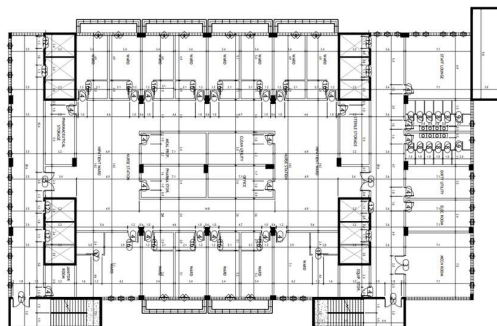
#### 3.1 Spatial Configuration & Architectural Data:

The simulation is based on a specific hospital ward layout designed to evaluate the acoustic environment of patient recovery zones. The architectural model encompasses:

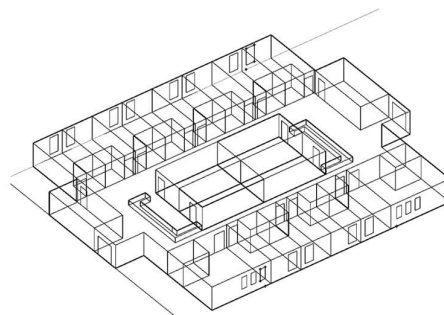
Total Volume: 2173.00 m<sup>3</sup>.

Clear Height: 2.8 meters, representing a standard clinical height that heavily influences early ceiling reflections.

Room Distribution: The ward consists of 12 mid-size patient rooms, 2 large high-occupancy wards, two central nurse stations serving as clinical hubs, and 4 centrally located patient service rooms for meal prep and utility.



**Figure 4.** Plan.



**Figure 5.**Ward zone area 3d.

### 3.2 Materials:

|                                                       |         |         |         |         |          |         |         |         |
|-------------------------------------------------------|---------|---------|---------|---------|----------|---------|---------|---------|
|                                                       | 57      | 4002    | 0.050   | 0.000   | Normal ▼ | 1-WALLS |         |         |
| Mat. 4002: Painted plaster surface (Kristensen, 1984) |         |         |         |         |          |         |         |         |
|                                                       | 63 Hz   | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz  | 2000 Hz | 4000 Hz | 8000 Hz |
|                                                       | 0.02000 | 0.02000 | 0.02000 | 0.02000 | 0.02000  | 0.02000 | 0.02000 | 0.02000 |

Figure 6. Walls material.

|                                                                                              |         |         |         |         |         |                   |         |         |
|----------------------------------------------------------------------------------------------|---------|---------|---------|---------|---------|-------------------|---------|---------|
|                                                                                              | 102     | 10100   | 0.050   | 0.000   | Normal  | ▼ 3-NURSE_STATION |         |         |
| Mat. 10100: (Example 10100-10199) S-LINE Office screen. wood Strict Line (href= "http://www. |         |         |         |         |         |                   |         |         |
|             | 63 Hz   | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz | 2000 Hz           | 4000 Hz | 8000 Hz |
|                                                                                              | 0.02000 | 0.02000 | 0.20000 | 0.70000 | 1.00000 | 0.90000           | 0.70000 | 0.70000 |

Figure 7. Partition material.

|                                                                                     |         |         |         |         |          |         |         |         |
|-------------------------------------------------------------------------------------|---------|---------|---------|---------|----------|---------|---------|---------|
|    | 59      | 12200   | 0.050   | 0.000   | Normal ▼ | CEILING |         |         |
| Mat. 12200: (Example 12200 12299) Ecophon (href= "http://www.ecophon.com/")         |         |         |         |         |          |         |         |         |
|  | 63 Hz   | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz  | 2000 Hz | 4000 Hz | 8000 Hz |
|                                                                                     | 0.15000 | 0.15000 | 0.50000 | 0.95000 | 1.00000  | 0.95000 | 0.85000 | 0.65000 |

Figure 8. Ceiling material.

|                                                                 |         |         |         |          |         |         |         |
|-----------------------------------------------------------------|---------|---------|---------|----------|---------|---------|---------|
| 84                                                              | 6000    | 0.050   | 0.000   | Normal ▼ | FLOOR   |         |         |
| Mat. 6000: Linoleum or vinyl stuck to concrete (Petersen, 1983) |         |         |         |          |         |         |         |
| 63 Hz                                                           | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz  | 2000 Hz | 4000 Hz | 8000 Hz |
| 0.02000                                                         | 0.02000 | 0.02000 | 0.03000 | 0.04000  | 0.04000 | 0.05000 | 0.05000 |

Figure 9. Floor material.

|                                                                        |         |         |         |         |          |         |         |         |
|------------------------------------------------------------------------|---------|---------|---------|---------|----------|---------|---------|---------|
|                                                                        | 121     | 10003   | 0.050   | 0.000   | Normal ▼ | WINDOWS |         |         |
| Mat. 10003: Double glazing, 2-3 mm glass, 10 mm gap (Kristensen, 1984) |         |         |         |         |          |         |         |         |
|                                                                        | 63 Hz   | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz  | 2000 Hz | 4000 Hz | 8000 Hz |
|                                                                        | 0.10000 | 0.10000 | 0.07000 | 0.05000 | 0.03000  | 0.02000 | 0.02000 | 0.02000 |

Figure 10. Windows material.

|                     |         |         |         |         |         |         |         |
|---------------------|---------|---------|---------|---------|---------|---------|---------|
| 77                  | 10007   | 0.050   | 0.000   | Normal  | ▼       | 2-DOORS |         |
| Mat. 0: Transperant |         |         |         |         |         |         |         |
| 63 Hz               | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz | 2000 Hz | 4000 Hz | 8000 Hz |
| 0.00000             | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 |

Figure 11. Interior door material.

|                     |         |         |         |         |         |         |         |
|---------------------|---------|---------|---------|---------|---------|---------|---------|
| 81                  | 0       | 0.050   | 0.000   | Normal  | ▼       | 2-DOORS |         |
| Mat. 0: Transperant |         |         |         |         |         |         |         |
| 63 Hz               | 125 Hz  | 250 Hz  | 500 Hz  | 1000 Hz | 2000 Hz | 4000 Hz | 8000 Hz |
| 0.00000             | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 | 0.00000 |

Figure 12. Exterior door material.

### 3.3 technical parameters & simulation setup:

The acoustical modeling was conducted using ODEON Room Acoustics Software, utilizing 48,359 rays for precision tracking. The primary technical parameter analyzed for global echo was the Reverberation Time (T30). The background noise floor was established by applying the standard NC 30 curve **Figure 13** to simulate a high-performance HVAC environment.

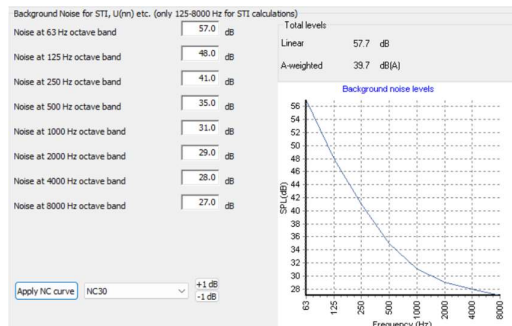


Figure 13. NC 30 curve.

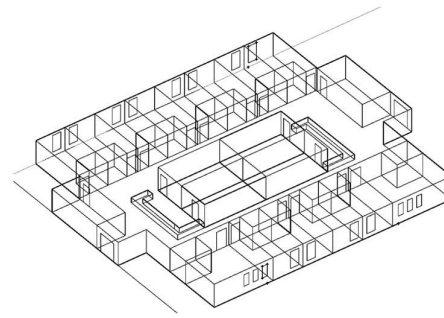
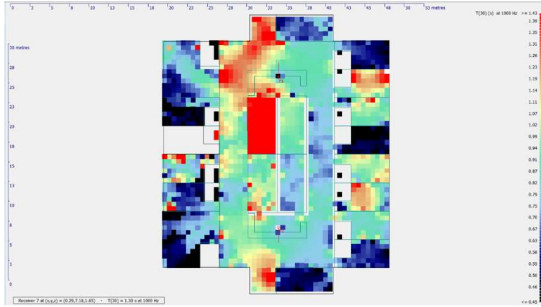


Figure 5. Ward zone area 3d.

### 3.4 T30 Extraction and Targeted Receiver Analysis:

To extract the relevant data and diagnose acoustic failures, a two-step analysis method was employed:

**Global Grid Mapping:** First, a global receiver grid was established at a seated patient height of 1.65 m. The T30 was calculated across the entire ward specifically at the 1000 Hz frequency band under the NC 30 scenario. This global heatmap revealed distinct "red zones"—localized areas where the reverberation time spiked excessively (exceeding 1.30 s) despite the presence of an acoustic ceiling. As shown in **Figure 14**.



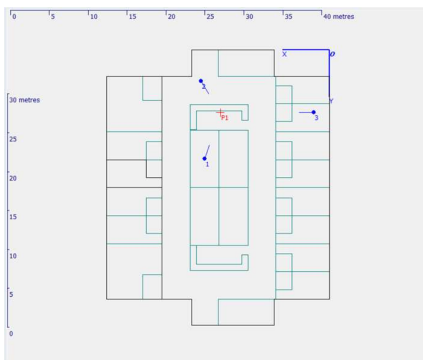
**Figure 14.**  $T_{30}$  at 1000Hz results.

**Single-Point Diagnostics:** To investigate the specific reflection paths causing these red zones, three targeted single-point receivers were placed directly into the most problematic areas shown in **Figure 15** identified on the map:

Head Nurse Office Receiver (Position:  $x=16.00$ ,  $y=14.00$ ,  $z=1.65$ ).

Corridor Receiver (Position:  $x=16.50$ ,  $y=4.00$ ,  $z=1.65$ ).

Patient Room Receiver (Position:  $x=2.00$ ,  $y=8.00$ ,  $z=1.65$ ).



**Figure 15.** Source & receiver location plan.

**Job Configuration:** Within ODEON, individual calculation jobs (Jobs 2, 3, and 4) were created for each of these three receivers. To isolate how noise from the primary clinical hub propagates into these specific zones, all three single-point receivers were explicitly pointed towards Source 1 (Nurse Station 1, located at  $x=14.00$ ,  $y=8.00$ ,  $z=1.65$ ). This setup allowed for the generation of precise reflectograms to track the exact multi-bounce paths contributing to the high  $T_{30}$  values at each location.



## 4. Data Analysis & Results:

This analysis evaluates the acoustic performance of a hospital ward based on data from three receiver points. The study focuses on **Reverberation Time T30** and **spatial reflections** to assess clinical suitability and speech intelligibility.

### 4.1 Receiver 1:

#### 4.1.1 Reverberation Time (T30) Frequency Distribution:

- **Low Frequency Performance:** The results show excessively high reverberation at 63 Hz and 125 Hz, peaking at approximately 2.4 seconds. This indicates a significant "boomy" effect caused by the lack of low-frequency absorption in the large 2173.00 m<sup>3</sup> ward volume.
- **Speech Frequencies:** At the critical 1000 Hz band, the T30 is measured at approximately 1.18 seconds. This confirms the "Red Zone" as shown in **Figure 14** data from the heatmap, showing that even at mid-frequencies, the reverberation remains nearly triple the preferred level for a patient-care environment.
- **High Frequency Decay:** There is a natural roll-off at 4000 Hz and 8000 Hz, dropping below 1.0 second, likely due to air absorption and the minimal scattering provided by the ward's internal geometry.
- **Overall Acoustic Condition:** The data confirms that Receiver 1 is in a highly reflective environment where late energy persists across all frequencies, particularly in the bass and mid-range, which will severely degrade speech intelligibility.

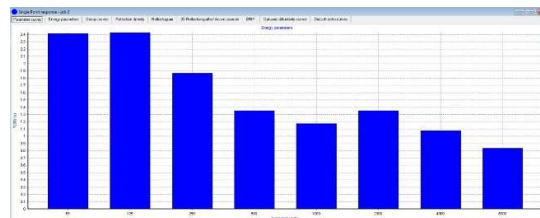


Figure 16. Reverberation Time (T30) Frequency Distribution for Receiver 1.

#### 4.1.2 Reflectogram and Spatial Reflection Analysis:

##### General Condition of the Reflectogram:

The Reflectogram in **Figure 7**, visualizes the sound energy arriving at the receiver over a timeline of approximately 0.22 seconds. The graph displays a dense field of high-amplitude spikes (reflections), indicating that the ward's surfaces are not providing sufficient absorption. The integrated red decay curve shows a gradual, slow slope rather than a sharp drop, visually confirming the over-reverberant state of the ward and the high T30 values that exceed the NC 30 requirements.



Figure 17. Reflectogram Analysis for Receiver 1.

### point 1 Analysis in Reflectogram:

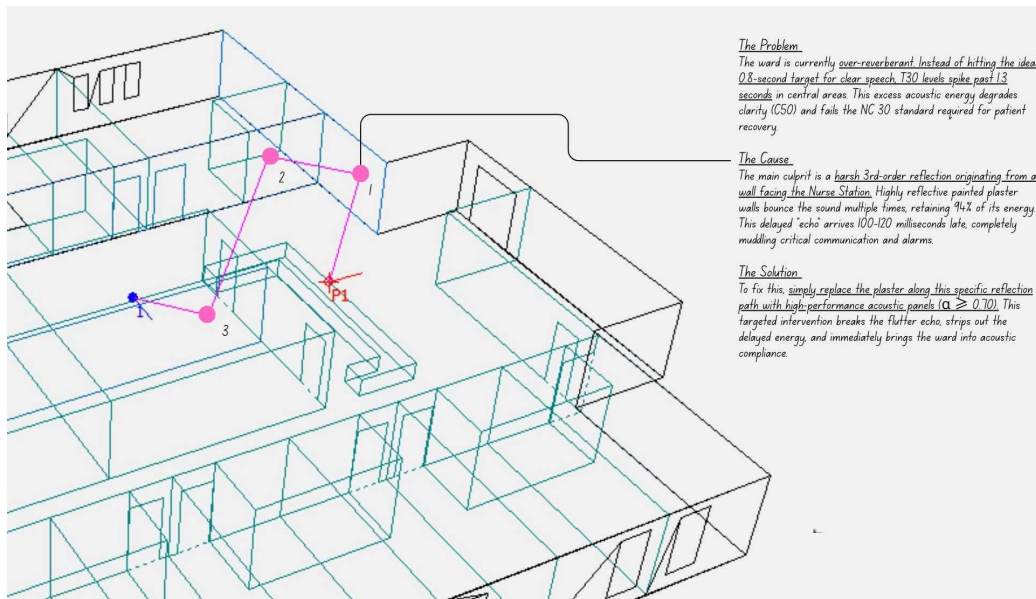


Figure 18. 3D Reflected Path At Point 1.

### point 2 Analysis in Reflectogram:

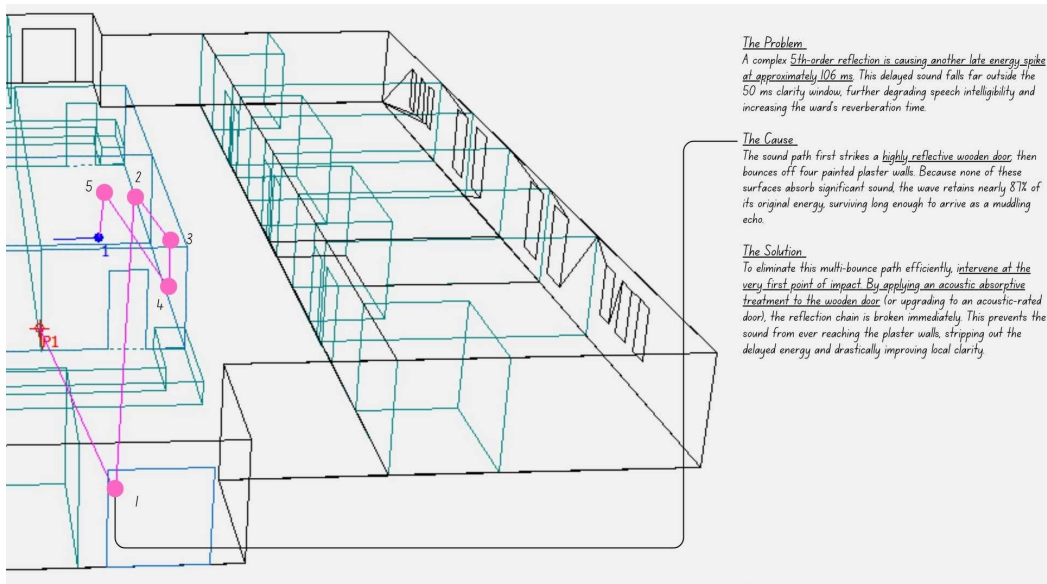


Figure 19. 3D Reflected Path At Point 2.

### point 3 Analysis in Reflectogram:

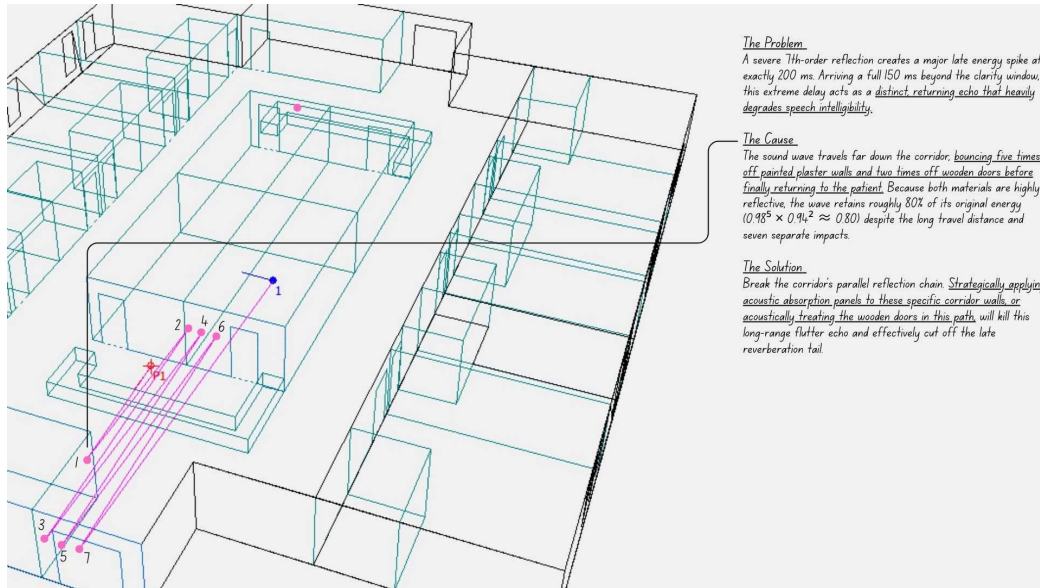


Figure 20. 3D Reflected Path At Point 3.

## 4.2 Receiver 2:

### 4.2.1 Reverberation Time (T30) Frequency Distribution:

- **Low-Frequency Performance:** The ward exhibits extremely high reverberation in the bass frequencies, with values exceeding **2.1 seconds** at **63 Hz** and **125 Hz**. This confirms a significant "boomy" effect caused by the 2173.00 m<sup>3</sup> volume and lack of low-frequency absorption.
- **Speech Frequency Range:** At the critical **1000 Hz** mid-frequency band, the T(30) is approximately **0.84 seconds**. While this is lower than the central core average, it still exceeds the ideal target for high-clarity clinical environments.
- **High-Frequency Decay:** A steady decrease is observed from **2000 Hz** down to **8000 Hz**, where the reverberation time drops to approximately **0.7 seconds**. This is largely due to natural air absorption and the limited energy return of high-pitched sounds.
- **Acoustic Summary:** The data reveals a "tilt" toward low-frequency energy. This spectral imbalance means that low-frequency background noise (like HVAC or distant footfalls) will linger much longer than speech, potentially masking important verbal communication.

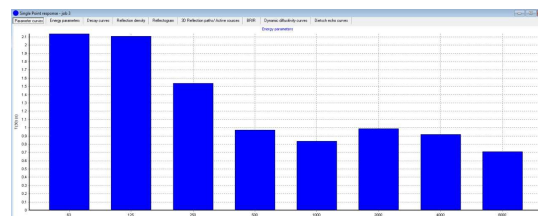


Figure 21. Reverberation Time (T30) Frequency Distribution for Receiver 2.

## 4.2.2 Reflectogram and Spatial Reflection Analysis:

### General Condition of the Reflectogram:

The Reflectogram in **Figure22** visualizes the sound energy arriving at the receiver over a timeline of approximately **0.33 seconds**. The graph displays a dense field of high-amplitude spikes (reflections), indicating that the ward's surfaces are not providing sufficient absorption. The integrated red decay curve shows a gradual, slow slope rather than a sharp drop, visually confirming the over-reverberant state of the ward and the high **T30** values that exceed the **NC 30** requirements.

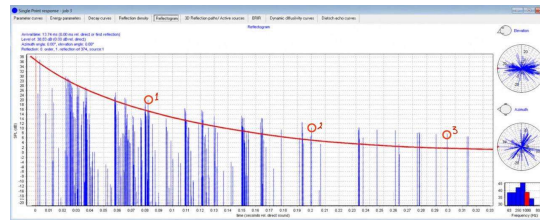


Figure 22. Reflectogram Analysis for Receiver 2.

### point 1 Analysis in Reflectogram:

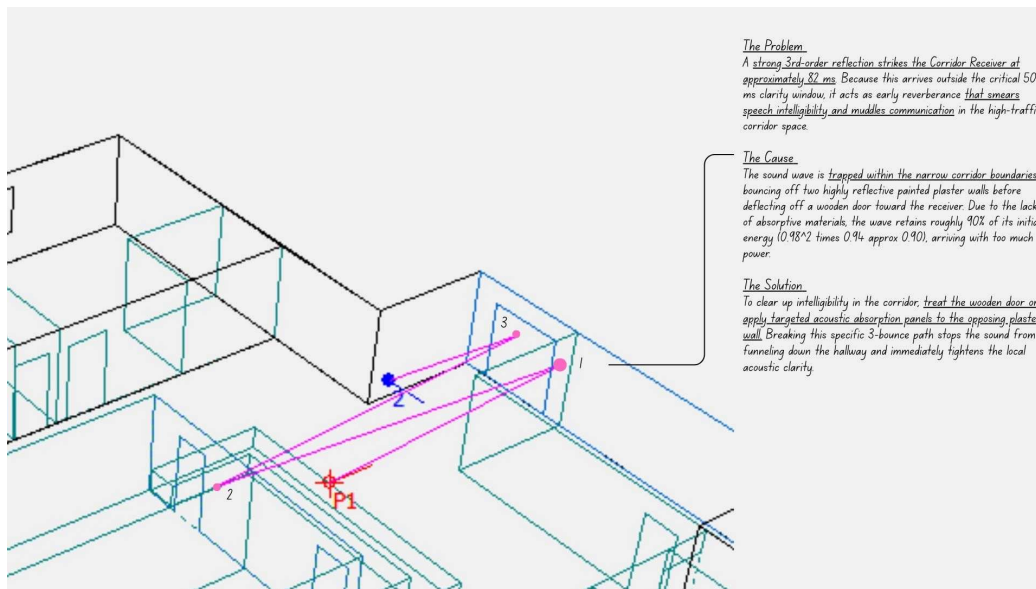


Figure 23. 3D Reflected Path At Point 1.



## Point 2 Analysis in Reflectogram:

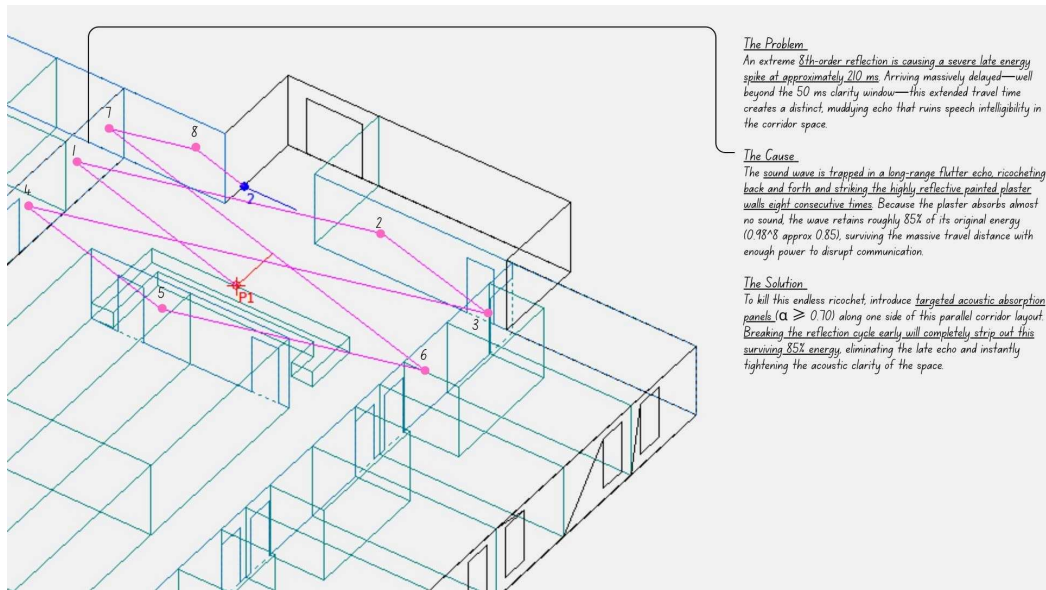


Figure 24. 3D Reflected Path At Point 2.

## Point 3 Analysis in Reflectogram:

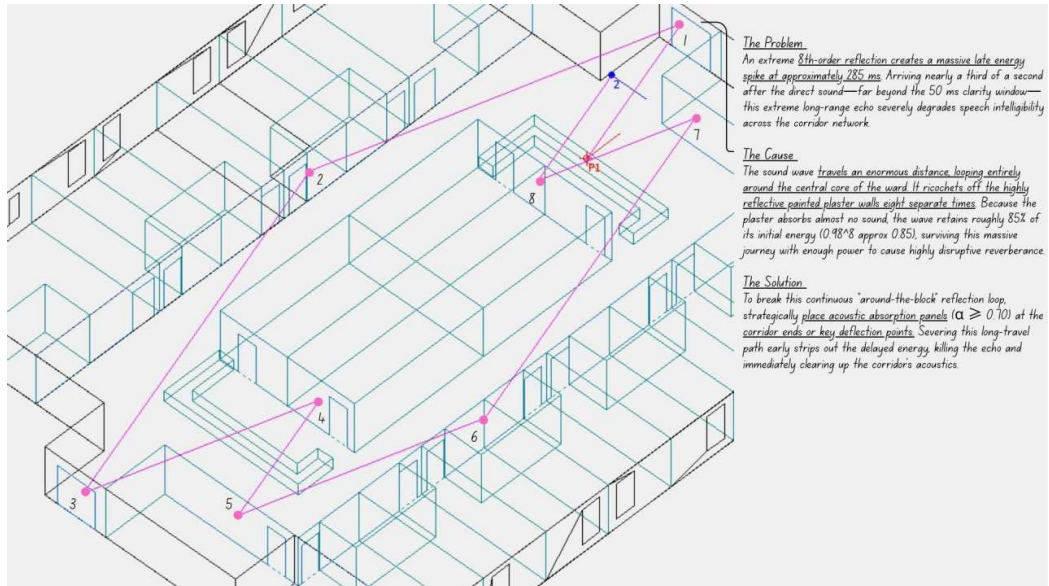
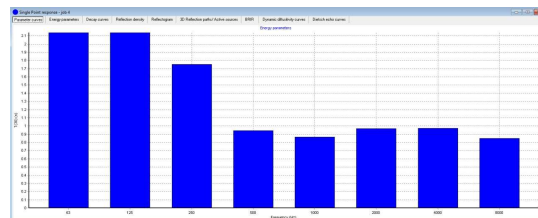


Figure 25. 3D Reflected Path At Point 3.

### 4.3 Receiver 3:

#### 4.3.1 Reverberation Time (T30) Frequency Distribution:

- **Low-Frequency Performance:** The ward continues to show excessively high reverberation in the bass frequencies, peaking at over **2.1 seconds** at **63 Hz** and **125 Hz**. This confirms a persistent "boomy" effect caused by the large volume and reflective surfaces.
- **Speech Frequency Range:** At the critical **1000 Hz** mid-frequency band, the T(30) is approximately **0.86 seconds**. While this is an improvement over the low-frequency peaks, it remains above the ideal target for clear clinical communication.
- **High-Frequency Decay:** The reverberation time stays relatively consistent between **500 Hz** and **4000 Hz** (ranging from **0.86s to 0.98s**) before dropping slightly at **8000 Hz** to **0.85 seconds**.
- **Acoustic Summary:** The data reveals a significant spectral imbalance where low-frequency sounds linger more than twice as long as mid-range speech frequencies. This imbalance will likely result in low-frequency background noise masking critical verbal instructions and medical alarms.

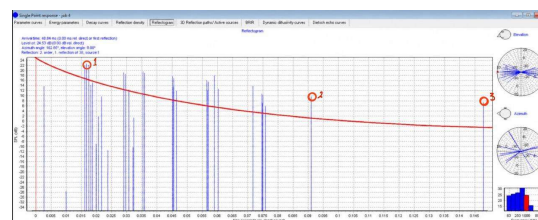


**Figure 26.** Reverberation Time (T30) Frequency Distribution for Receiver 3.

#### 4.3.2 Reflectogram and Spatial Reflection Analysis:

##### General Condition of the Reflectogram:

The Reflectogram in **Figure 27** visualizes the sound energy arriving at the receiver over a timeline of approximately **0.15 seconds**. The graph displays a dense field of high-amplitude spikes (reflections), indicating that the ward's surfaces are not providing sufficient absorption. The integrated red decay curve shows a gradual, slow slope rather than a sharp drop, visually confirming the over-reverberant state of the ward and the high **T30** values that exceed the **NC 30** requirements.



**Figure 27.** Reflectogram Analysis for Receiver 3.

### point 1 Analysis in Reflectogram:

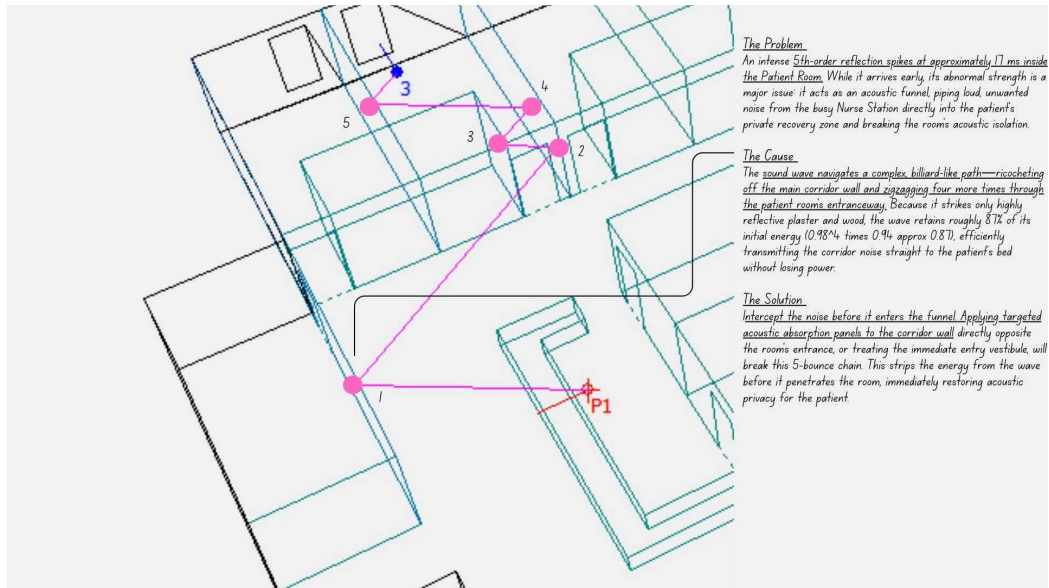


Figure 28. 3D Reflected Path At Point 1.

### Point 2 Analysis in Reflectogram:

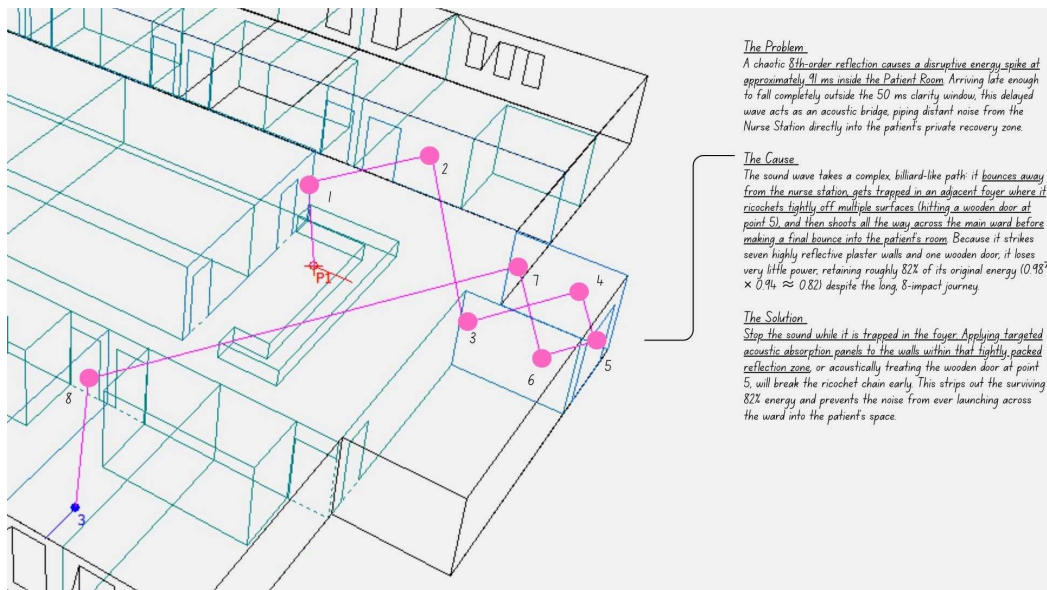


Figure 29. 3D Reflected Path At Point 2.

### Point 3 Analysis in Reflectogram:

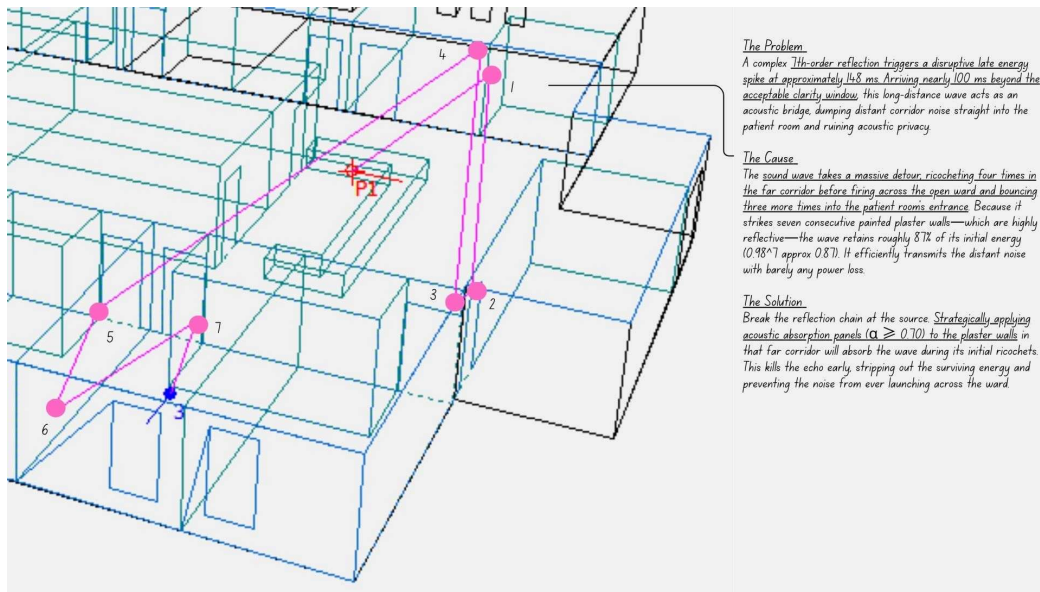


Figure 30. 3D Reflected Path At Point 3.



## 5. Conclusion

The acoustic analysis of the hospital ward reveals a significant departure from the standards required for a clinical environment. While the three receiver points represent different spatial zones, they share a common failure: **excessive reverberation and poor speech intelligibility** caused by highly reflective internal surfaces.

### 5.1 Summary of Key Findings

- **Acoustic Over-Reverberance:** Across all receiver locations, the Reverberation Time T30 consistently exceeds the ideal target of **0.8 seconds**. Central areas (Receiver 1) are particularly problematic, with mid-frequency decay reaching nearly **1.2 seconds**—triple the preferred level for patient care.
- **Low-Frequency "Boom" Effect:** The ward suffers from a severe spectral imbalance. Low frequencies Hz125 peak at over **2.1 to 2.4 seconds**, creating a "boomy" environment where background noise from HVAC and footfalls can easily mask critical medical alarms and verbal instructions.
- **Pathological Reflections:** The 3D reflection analysis identifies specific "culprit" paths—ranging from 3rd-order to 8th-order reflections—where sound ricochets off painted plaster walls and wooden doors. These paths retain up to **94% of their energy**, resulting in distinct echoes that arrive well outside the 50 ms clarity window.

### 5.2 Strategic Recommendations

To bring the facility into compliance with **NC 30** standards and ensure a safe, therapeutic environment, the following interventions are recommended:

1. **Targeted Absorption:** Replace existing painted plaster at key impact points with high-performance acoustic panels 0.70.
2. **Corridor Treatment:** Break the parallel reflection chains in the corridors by applying absorptive materials to one side of the hallway or upgrading wooden doors to acoustic-rated alternatives.
3. **Low-Frequency Control:** Incorporate bass-trapping elements or perforated acoustic ceilings to dampen the excessive low-frequency energy caused by the ward's large 2173.00 volume.

By implementing these targeted acoustic treatments, the ward can transition from its current "Red Zone" status to a compliant, high-clarity environment that supports both patient recovery and clear clinical communication.

## References

1. Barron, M. (2009) *Auditorium Acoustics and Architectural Design*. 2nd edn. London: Spon Press.
2. Beranek, L. and Mellow, T. (2012) *Acoustics: Sound Fields and Transducers*. Oxford: Academic Press.
3. Cox, T.J. and D'Antonio, P. (2016) *Acoustic Absorbers and Diffusers: Theory, Design and Application*. 3rd edn. Boca Raton: CRC Press.
4. Department of Health (2013) *Health Technical Memorandum 08-01: Acoustics*. London: TSO.
5. Everest, F.A. and Pohlmann, K.C. (2014) *Master Handbook of Acoustics*. 6th edn. New York: McGraw-Hill.
6. ISO 3382-1 (2009) *Acoustics — Measurement of room acoustic parameters — Part 1: Performance spaces*. Geneva: International Organization for Standardization.
7. Kuttruff, H. (2016) *Room Acoustics*. 6th edn. Boca Raton: CRC Press.
8. Long, M. (2014) *Architectural Acoustics*. 2nd edn. Oxford: Elsevier.